Basic Java Programs

1. Java Program to Check Leap Year

```
public class LeapYear {
    public static void main(String[] args) {
        int year = 2024;

        boolean isLeap = (year % 4 == 0 && year % 100 != 0) || (year % 400
== 0);

        if (isLeap) {
            System.out.println(year + " is a leap year.");
        } else {
            System.out.println(year + " is not a leap year.");
        }
    }
}
```

2. Java Program to Check Whether a Character is Alphabet or Not

```
public class CheckAlphabet {
    public static void main(String[] args) {
        char ch = 'A';

        if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z')) {
            System.out.println(ch + " is an alphabet.");
        } else {
            System.out.println(ch + " is not an alphabet.");
        }
    }
}
```

3. Java Program to Find Factorial of a Number

```
public class Factorial {
    public static void main(String[] args) {
        int num = 5;
        long factorial = 1;

        for (int i = 1; i <= num; i++) {
            factorial *= i;
        }

        System.out.println("Factorial of " + num + " = " + factorial);
    }
}
```

4. Java Program to Display Fibonacci Series

```
public class Fibonacci {
    public static void main(String[] args) {
        int n = 10; // Number of terms
        int firstTerm = 0, secondTerm = 1;

        System.out.print("Fibonacci Series (" + n + " terms): ");
        for (int i = 1; i <= n; i++) {
            System.out.print(firstTerm + " ");

            int nextTerm = firstTerm + secondTerm;
            firstTerm = secondTerm;
            secondTerm = nextTerm;
        }
    }
}
```

5. Java Program to Find GCD of two Numbers

```
public class GCD {
    public static void main(String[] args) {
        int n1 = 81, n2 = 153;
        int a = n1, b = n2;

        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }

        System.out.println("GCD of " + n1 + " and " + n2 + " is: " + a);
    }
}
```

6. Java Program to Find LCM of two Numbers

```
public class LCM {
    public static void main(String[] args) {
        int n1 = 12, n2 = 18;
        int a = n1, b = n2;

        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }

        int gcd = a;
        int lcm = (n1 * n2) / gcd;

        System.out.println("LCM of " + n1 + " and " + n2 + " is: " + lcm);
    }
```

```
}
```

7. Java Program to Count Number of Digits in an Integer

```
public class CountDigits {
    public static void main(String[] args) {
        int num = 123456;
        int count = 0;
        int temp = Math.abs(num);

        if (temp == 0) {
            count = 1;
        } else {
            while (temp > 0) {
                temp /= 10;
                count++;
            }
        }

        System.out.println("Number of digits in " + num + " = " + count);
    }
}
```

8. Java Program to Reverse a Number

```
public class ReverseNumber {
    public static void main(String[] args) {
        int num = 1234;
        int reversed = 0;
        int temp = num;

        while (temp != 0) {
            int digit = temp % 10;
            reversed = reversed * 10 + digit;
            temp /= 10;
        }

        System.out.println("Original: " + num + ", Reversed: " +
reversed);
    }
}
```

9. Java Program to Calculate the Power of a Number

```
public class PowerCalculation {
    public static void main(String[] args) {
        double base = 2;
        int exponent = 5;
        double result = 1;
```

```
        for (int i = 1; i <= Math.abs(exponent); i++) {
            result *= base;
        }

        if (exponent < 0) {
            result = 1.0 / result;
        }

        System.out.println(base + "^" + exponent + " = " + result);
    }
}
```

10. Java Program to Check Palindrome

```
public class CheckPalindrome {
    public static void main(String[] args) {
        int num = 1221;
        int originalNum = num;
        int reversed = 0;

        while (num != 0) {
            int digit = num % 10;
            reversed = reversed * 10 + digit;
            num /= 10;
        }

        if (originalNum == reversed) {
            System.out.println(originalNum + " is a palindrome.");
        } else {
            System.out.println(originalNum + " is not a palindrome.");
        }
    }
}
```